

CENG381 Stochastic Processes

Wald's Equality and De Moivre's Martingale - Question Set 1

Notation used in this question set:

- Wald's Equality:** Let $X_1, X_2, \dots$ be i.i.d. with $E[X] = \mu$. Let T be a stopping time with $E[T] < \infty$, independent of future X_i . Then $E[S(T)] = \mu * E[T]$, where $S(T) = X_1 + X_2 + \dots + X_T$.
- Second Wald Identity:** If additionally $E[X] = 0$ and $\text{Var}[X] = \sigma^2$, then $E[S(T)^2] = \sigma^2 * E[T]$. (Equivalently, $S(n)^2 - \sigma^2 n$ is a martingale.)
- Biased random walk:** Steps X_i are $+1$ with probability p and -1 with probability $q = 1-p$, where $p \neq 1/2$. The walk has drift $E[X] = p - q = 2p - 1$.
- De Moivre's martingale:** For a biased walk with ratio $r = q/p$, the process $M(n) = r^{S(n)}$ is a martingale. This is verified by showing $E[r^{X_{n+1}}] = 1$, which gives $E[M(n+1) | \mathcal{F}_n] = M(n) * E[r^{X_{n+1}}] = M(n)$.
- Jensen's Inequality:** For a CONVEX function f and a random variable X with $E[|X|] < \infty$, $f(E[X]) \leq E[f(X)]$. Convex means the function curves upward: $f''(x) \geq 0$ for all x . Examples: $f(x) = x^2$, $f(x) = e^x$, $f(x) = r^x$ for $r > 1$.

De Moivre's martingale gives the Gambler's Ruin probability for the biased walk: apply OST to $r^{S(n)}$ with absorbing barriers at 0 and N to get $P(\text{reach } N | \text{start } k) = (1 - r^k) / (1 - r^N)$ when $r = q/p \neq 1$.

Q1 (30 points). You roll a fair die repeatedly. Let X_i be the outcome of roll i (X_i in $\{1,2,3,4,5,6\}$, each equally likely). Define $S(n) = X_1 + \dots + X_n$ and let $T = \min\{n : S(n) \geq 100\}$ (stop as soon as the sum reaches 100).

- Compute $E[X_i]$ and $\text{Var}[X_i]$ for a fair die. (Recall: $E[X] = (1+2+\dots+6)/6$ and $E[X^2] = (1^2+\dots+6^2)/6$.)
- Use Wald's Equality $E[S(T)] = E[X] * E[T]$ to estimate $E[T]$. Note that when the walk hits 100 it may overshoot slightly ($S(T) \geq 100$ but $S(T-1) < 100$), so $S(T) \sim 100$ approximately. Give the approximate value of $E[T]$.
- A casino game pays out \$1 if the player reaches exactly sum = 100 with their rolls, and \$0 otherwise. Would you expect this to be a common or rare event? Using the fact that the probability of hitting exactly 100 (rather than overshooting) is approximately $1/E[X] =$

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2/7 by the renewal theory result, what is the approximate probability of landing on exactly 100?

Q2 (35 points). A biased random walk has $P(X_i = +1) = p = 3/5$ and $P(X_i = -1) = q = 2/5$. The walk starts at position $k = 2$ with absorbing barriers at 0 (ruin) and $N = 6$ (goal).

a) Compute the ratio $r = q/p$. Verify that $M(n) = r^{S(n)}$ is a martingale by computing:

$$E[r^{X_{n+1}}] = p * r^{+1} + q * r^{-1}$$

and showing it equals 1.

b) Apply the Optional Stopping Theorem to $M(n) = r^{S(n)}$ with $T =$ first hitting time of $\{0, N\}$. Let $h = P(\text{reach } N=6 \text{ before } 0 \mid \text{start } k=2)$. Set up and solve:

$$E[M(T)] = E[M(0)]$$

$$r^6 * h + r^0 * (1-h) = r^2$$

Solve for h .

c) Compare the biased walk result h from (b) to the symmetric walk result

$h_{\text{sym}} = k/N = 2/6 = 1/3$. Is the gambler more or less likely to reach

the goal with the biased walk ($p=3/5 > 1/2$)? Explain intuitively.

Also compute $P(\text{ruin before goal}) = 1 - h$.

Q3 (35 points). An asset price evolves as a random multiplicative process: at each time step the price is multiplied by either $u = 1.1$ (up) with probability p or $d = 0.9$ (down) with probability $1-p$. Starting price is $S(0) = 100$.

a) Write $S(n) = 100 * u^{A_n} * d^{B_n}$ where A_n is the number of up moves and $B_n = n - A_n$ is the number of down moves in n steps. Under what probability p does $E[S(n)] = S(0)$ for all n ? (This is the 'risk-neutral' probability. Set up $E[S(1)] = S(0)$ and solve.)

b) Under the risk-neutral probability from (a), show that $S(n)$ is a martingale by verifying $E[S(n+1) \mid \mathcal{F}_n] = S(n)$. (Write $S(n+1) = S(n) * Y_{n+1}$ where $Y_{n+1} = u$ with prob p or d with prob $1-p$, and compute $E[Y_{n+1}]$.)

c) By Jensen's Inequality applied to $f(x) = e^x$ (which is convex), we have $E[e^X] \geq e^{E[X]}$ for any random variable X . Use this to argue that for $p >$ risk-neutral probability (the real-world drift is upward), we have $E[S(n)] > S(0)$ under the real-world measure. Is $S(n)$ then a martingale, supermartingale, or submartingale under the real-world measure when $p > p_{\text{risk_neutral}}$?